

SONU DIXIT

☎ 83750-68648 ✉ sonudixit2k@gmail.com 🔗 [linkedin.com/in/sonudixit](https://www.linkedin.com/in/sonudixit)

Summary

Machine learning practitioner with experience in NLP, Information Retrieval, LLM post-training. Currently working on reasoning-based content moderation using SFT and RLVR for multimodal foundation models.

Experience

LinkedIn

Apr 2025 - Present

Reasoning-Based Content Moderation

Oct 2025 - Present

- Built an end-to-end SFT + RLVR post-training pipeline for open-weight multimodal LLMs to perform policy-based content moderation.
- Improved moderation recall by 2× over GPT-based moderation systems while maintaining adaptability to evolving policy definitions.
- Designed evaluation frameworks for reasoning quality, instruction following, and robustness under unseen moderation policies.

[24]7.ai, Data Scientist-2

Feb 22 - Oct 24

LLM Finetuning for Customer Support Conversations

Aug 2024 - Oct 2024

- Fine-tuned Llama-3.1-8B-Instruct on customer-agent conversations to improve response style, personalization, and conversational quality.
- Studied the impact of task diversity during post-training and identified degradation in general instruction-following when training exclusively on dialogue data.

Intent Classification

Dec 2023 - Sep 2024

- Developed transformer-based intent classification models for customer-support automation, outperforming production systems, GPT-3.5 few-shot prompting, and SetFit baselines.
- Designed experiments to evaluate robustness under out-of-vocabulary terminology, distribution shifts, and evolving intent taxonomies.
- Investigated model bias and developed mitigation strategies to improve reliability across client deployments.

Knowledge Article Recommendation

Feb 2022 - Dec 2022

- Trained dense retrieval models including BERT-dot, BERT-cat, and ColBERT for passage ranking on MS MARCO.
- Achieved competitive performance on TREC Deep Learning benchmarks using a distilled DistilBERT retriever.
- Developed a dual-distillation strategy using BERT-cat and ColBERT teacher models to reduce inference cost while preserving retrieval quality.

Disney Star India – Assistant Manager (Data Science)

Jul 2019 – Jan 2022

- Built storyline extraction pipelines using representation learning, clustering, summarization, and relationship graph construction.
- Developed HMM-based sequence models for extracting entities from semi-structured business documents.

Publications

- Zero-Shot Generalization using Intrinsically Motivated Compositional Emergent Protocols, R Hazra*, [Sonu Dixit*](#), S Sen in *NAACL 2021 workshop - ViGIL*
- gComm: An environment for investigating generalization in Grounded Language Acquisition, R Hazra, [Sonu Dixit](#) in *NAACL 2021 workshop - ViGIL*

Education

Indian Institute of Science

2017 – 2019

MTech (Artificial Intelligence)

Overall GPA: 7.5/10

Gurukula Kangri University

2012 – 2016

BTech (Computer Science)

Overall percentage: 78.60

Projects

Adaptive Traffic Signal Control | *Multi Agent RL, PPO*

MTech Thesis

- Implemented multi-agent reinforcement learning (MARL) to adjust traffic signal durations based on congestion dynamically. Algorithm - Proximal Policy Optimization(PPO) with Advantage Actor-Critic
- Simulated traffic data using PTV Vissim and demonstrated MARL's superiority over fixed-time algorithms in terms of average speed, delay, and lane occupancy. [Report]